

Evaluation

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5.1 Evaluating Results

Model performance was assessed using the metrics defined in the Phase 4 test design. The table below presents the evaluation results for all three models on the held-out test set.

Model	MAE	MSE	RMSE	R ²	ASE	SSE	ObservedAvg
Random Forest	35.58	5978.4	77.32	0.946	5978.4	169913246.5	1016.17
HistGradientBoosting	47.07	7344.9	85.70	0.934	7344.9	208749725.6	1016.17
XGBoost	46.47	6820.1	82.58	0.939	6820.1	193833558.3	1016.17

Table 1

Model Performance Metrics Comparison

Random Forest achieved the highest R² of 0.946 and the lowest error metrics across all categories, indicating the strongest predictive performance among the three models. XGBoost ranked second with an R² of 0.939, while HistGradientBoostingRegressor achieved an R² of 0.934. All three models performed well, with the gap between the best and worst model being only 0.012 in R². For this project, these results mean that Random Forest is the strongest candidate for estimating rental prices from listing data. Its performance suggests that the model can produce predictions reasonably close to actual prices for many properties in the dataset, making it the most reliable of the three tested approaches. The Observed Average price of \$1,016.17 provides a meaningful baseline: Random Forest's MAE of \$35.58 represents an average prediction error of approximately 3.5% relative to the mean listing price, which is a strong result for this domain. This is important because the main goal of the project is to produce a predictive understanding of what drives apartment listing prices in the Southern United States.

The feature importance results confirm that location-based features are the dominant drivers of rental price. The location_cluster feature created by K-Means clustering on latitude and longitude coordinates ranked as the most important predictor across all three models. Region and square footage (sqfeet) consistently appeared in the top three. Table 2 summarises the top features and their rankings across all three models.

Feature	RF	HistGB	XGB
location_cluster	1	1	1
region	2	3	2
sqfeet	3	2	—
lat	4	—	—
long	—	4	—
baths	—	—	3
laundry_options	—	—	4

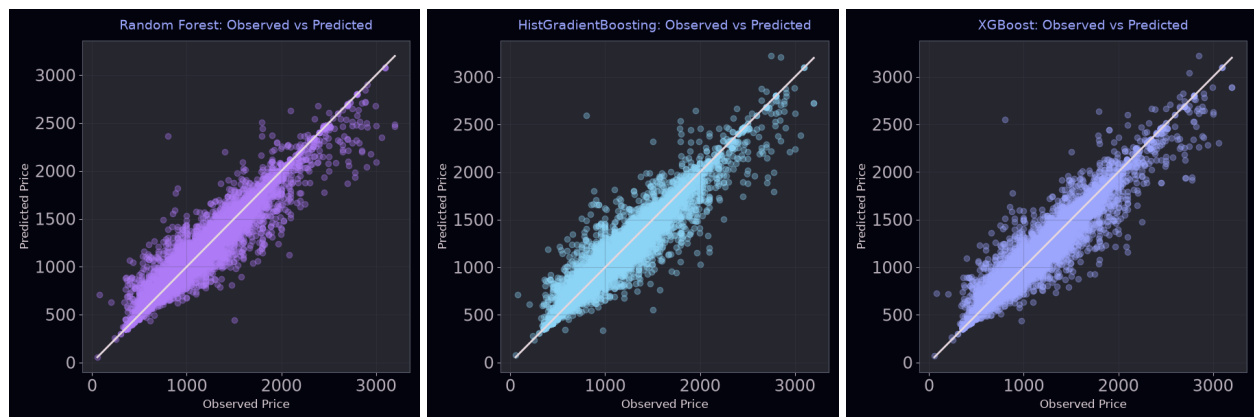
Table 2

Feature importance rankings across models (blank = outside top 4)

The cross-model agreement is notable: `location_cluster` is the most important predictor in every model, confirming that geographic clustering captured meaningful price variation that individual region labels alone could not. For renters, this means the model can help assess whether a property is reasonably priced relative to its area and size. For landlords and property managers, it shows which factors have the greatest influence on expected rental value, and that individual amenities such as laundry options or pet policies matter far less than location and size.

Figure 1

Observed vs Predicted Values

(a) *Random Forest*(b) *HistGradientBoosting*(c) *XGBoost*

The observed-versus-predicted plots (Figure 1) further support the usefulness of the modeling approach, since most predictions follow the general trend of the actual values. However, a slight broadening of the spread at higher price levels suggests that the models are less precise for the most expensive listings. This indicates the project shows strong promise, but is not yet ready for

deployment without further tuning and validation.

5.2 Reviewing the Process

The process review captures several important lessons from the project.

First, the initial modeling iterations did not use K-Means for location clustering, and the results were noticeably weaker. Adding location clustering around Craigslist-defined regions (`n_clusters=148`) substantially improved model performance. The `location_cluster` feature emerged as the number-one predictor across all three models, confirming that this engineering decision was one of the most impactful of the project.

Second, the project initially applied IQR-based outlier handling at the national level. After changing the procedure to apply IQR filtering by individual region, the models improved notably. This indicates that region-sensitive cleaning better reflected the structure of the housing market data, where price distributions vary meaningfully between different metropolitan areas.

5.3 Determining the Next Steps

The following potential improvements were identified for future iterations of this project:

- Optimize K-Means clustering by systematically finding the optimal number of clusters (e.g., elbow method or silhouette scores) rather than matching the number of Craigslist regions.
- Using newer and larger datasets would improve model relevancy to current market forces and expand geographic coverage beyond the Southern USA.
- Applying NLP to the description field could reveal insights missed by structured features, such as renovation quality or neighborhood character.
- Use GPU/CUDA to speed up training, tuning, and hyperparameter search for more extensive experimentation.
- Combining all three models in a weighted ensemble could improve prediction reliability by leveraging the strengths of each approach.
- Additional feature engineering (such as price-per-square-foot ratios, proximity to urban centers, or property age estimation) could capture further explanatory power.

On average, all three models performed well, and the modeling approach showed promising predictive performance. However, the project is not yet ready for deployment as-is. The final recommendation is to revise and improve the work before deployment, since the results are encouraging but further tuning, extension, and validation are still needed.